Grouped Permutation Feature Importance across Candidate Models and Physical Domains

Atmospheric Demand & Heat | Hydrological Supply | Soil Moisture Dynamics | Thermal & Radiation Regimes | Memory & Spatial Baseline

(a) ElasticNet (+Lag)

- Vapor Pressure Deficit (VPD): 10.5%
- Reference Evapotranspiration (ET0): 1.1%
- Moderate Heat Days (>=30C):
- Extreme Heat Days (>=35C): 0.3%
- Extreme Degree Days (>30C): 10.8%
- Cumulative Precipitation (P): 11.9%
- Climatic Water Balance (P - ET0): 2.8%
- Topsoil Moisture (0-7 cm): 10.0%
- Mid Subsoil Moisture (7-28 cm): 4.0%
- Deep Subsoil Moisture (28-100 cm): 3.1%
- Root-Zone Soil Moisture: 1.1%
- Mean Temperature (T_mean):
- Max Temperature (T_max): 5.6%
- Min Temperature (T_min): 1.7%
- Growing Degree Days (GDD): 0.4%
- Dewpoint Temperature (T_dew): 29.2%
- Surface Solar Radiation (SSRD): 5.9%
- Lagged Yield Memory (ε_{t-1}): 0.6%
- Spatial Coordinates (Lat / Lon): 1.0%

0% 5% 10% 15% 20% 25% 30%

Information Share (%)

(b) XGBoost (+Lag)

- Vapor Pressure Deficit (VPD): 15.9%
- Reference Evapotranspiration (ET0): 3.8%
- Moderate Heat Days (>=30C):
- Extreme Heat Days (>=35C):
- Extreme Degree Days (>30C):
- Cumulative Precipitation (P): 13.1%
- Climatic Water Balance (P - ET0): 6.2%
- Topsoil Moisture (0-7 cm): 6.5%
- Mid Subsoil Moisture (7-28 cm): 5.6%
- Deep Subsoil Moisture (28-100 cm): 0.7%
- Root-Zone Soil Moisture: 1.4%
- Mean Temperature (T_mean): 4.0%
- Max Temperature (T_max): 3.9%
- Min Temperature (T_min): 9.8%
- Growing Degree Days (GDD): 2.2%
- Dewpoint Temperature (T_dew): 15.2%
- Surface Solar Radiation (SSRD): 8.0%
- Lagged Yield Memory (ε_{t-1}): 3.7%
- Spatial Coordinates (Lat / Lon):

0% 5% 10% 15% 20% 25% 30%

Information Share (%)

(c) LSTM (Surrogate)

- Vapor Pressure Deficit (VPD): 7.7%
- Reference Evapotranspiration (ET0): 3.1%
- Moderate Heat Days (>=30C): 0.9%
- Extreme Heat Days (>=35C): 0.2%
- Extreme Degree Days (>30C): 3.1%
- Cumulative Precipitation (P): 43.4%
- Climatic Water Balance (P - ET0): 18.4%
- Topsoil Moisture (0-7 cm): 3.1%
- Mid Subsoil Moisture (7-28 cm): 1.1%
- Deep Subsoil Moisture (28-100 cm): 2.7%
- Root-Zone Soil Moisture:
- Mean Temperature (T_mean): 0.3%
- Max Temperature (T_max): 1.1%
- Min Temperature (T_min): 0.3%
- Growing Degree Days (GDD): 0.9%
- Dewpoint Temperature (T_dew): 10.6%
- Surface Solar Radiation (SSRD): 1.8%
- Lagged Yield Memory (ε_{t-1}): 1.1%
- Spatial Coordinates (Lat / Lon):

0% 5% 10% 15% 20% 25% 30%

Information Share (%)